

Supplementary Table 7. Cross-fitted recalibration metrics, recalibration differences, and decision curve summaries

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Standalone display version for LaTeX upload/review. Machine-readable source: `source_tables/Supplementary_Table_7_rec calibration_and_decision_curve.csv`. Rows: 126; columns: 15. The source table is split below into source-defined sections and column blocks so that all non-empty cells remain readable.

table block: Recalibration metrics

Table 1: Supplementary Table 7, Recalibration metrics, column block 1 of 2

model	calibration method	metric	point	bootstrap mean
Classical signal features	raw	auroc	0.6303	0.6293
Classical signal features	raw	auprc	0.3366	0.3522
Classical signal features	raw	brier	0.2525	0.2528
Classical signal features	raw	Calibration intercept	-1.384	-1.4
Classical signal features	raw	Calibration slope	0.4876	0.4923
Classical signal features	raw	Expected calibration error	0.2745	0.2745
Classical signal features	raw	Mean prediction error	0.2745	0.2735
Classical signal features	platt	auroc	0.6303	0.6293
Classical signal features	platt	auprc	0.3366	0.3522
Classical signal features	platt	brier	0.1585	0.1591
Classical signal features	platt	Calibration intercept	-0.4845	-0.4923
Classical signal features	platt	Calibration slope	0.6906	0.6972
Classical signal features	platt	Expected calibration error	0.0527	0.074
Classical signal features	platt	Mean prediction error	0.0212	0.0203
Classical signal features	isotonic	auroc	0.639	0.6382
Classical signal features	isotonic	auprc	0.3064	0.3175
Classical signal features	isotonic	brier	0.1659	0.1664
Classical signal features	isotonic	Calibration intercept	-0.716	-0.7232
Classical signal features	isotonic	Calibration slope	0.4983	0.5031
Classical signal features	isotonic	Expected calibration error	0.0625	0.0845
Classical signal features	isotonic	Mean prediction error	0.0237	0.0228
Signal + SSL	raw	auroc	0.6025	0.6007
Signal + SSL	raw	auprc	0.3765	0.3832
Signal + SSL	raw	brier	0.2701	0.2707
Signal + SSL	raw	Calibration intercept	-1.36	-1.377
Signal + SSL	raw	Calibration slope	0.3136	0.3132
Signal + SSL	raw	Expected calibration error	0.2601	0.2754
Signal + SSL	raw	Mean prediction error	0.2601	0.2593
Signal + SSL	platt	auroc	0.6025	0.6007
Signal + SSL	platt	auprc	0.3765	0.3832
Signal + SSL	platt	brier	0.1576	0.1583
Signal + SSL	platt	Calibration intercept	-0.1999	-0.2188
Signal + SSL	platt	Calibration slope	0.947	0.9458
Signal + SSL	platt	Expected calibration error	0.0491	0.0701
Signal + SSL	platt	Mean prediction error	0.0226	0.0216
Signal + SSL	isotonic	auroc	0.5916	0.5902
Signal + SSL	isotonic	auprc	0.254	0.2607
Signal + SSL	isotonic	brier	0.1658	0.1665
Signal + SSL	isotonic	Calibration intercept	-0.8159	-0.8166
Signal + SSL	isotonic	Calibration slope	0.4215	0.433
Signal + SSL	isotonic	Expected calibration error	0.0644	0.0772
Signal + SSL	isotonic	Mean prediction error	0.0236	0.0227

Continued from Supplementary Table 7, Recalibration metrics, column block 1 of 2.

model	calibration method	metric	point	bootstrap mean
Signal + SSL + phenotype	raw	auroc	0.6168	0.6151
Signal + SSL + phenotype	raw	auprc	0.4016	0.4069
Signal + SSL + phenotype	raw	brier	0.2652	0.2657
Signal + SSL + phenotype	raw	Calibration intercept	-1.359	-1.376
Signal + SSL + phenotype	raw	Calibration slope	0.3448	0.3461
Signal + SSL + phenotype	raw	Expected calibration error	0.2568	0.2699
Signal + SSL + phenotype	raw	Mean prediction error	0.2568	0.2562
Signal + SSL + phenotype	platt	auroc	0.6168	0.6151
Signal + SSL + phenotype	platt	auprc	0.4016	0.4069
Signal + SSL + phenotype	platt	brier	0.1566	0.1572
Signal + SSL + phenotype	platt	Calibration intercept	0.077	0.0641
Signal + SSL + phenotype	platt	Calibration slope	1.165	1.169
Signal + SSL + phenotype	platt	Expected calibration error	0.0322	0.0648
Signal + SSL + phenotype	platt	Mean prediction error	0.0192	0.0183
Signal + SSL + phenotype	isotonic	auroc	0.5896	0.5888
Signal + SSL + phenotype	isotonic	auprc	0.2672	0.2718
Signal + SSL + phenotype	isotonic	brier	0.1623	0.163
Signal + SSL + phenotype	isotonic	Calibration intercept	-0.6669	-0.6662
Signal + SSL + phenotype	isotonic	Calibration slope	0.5354	0.5495
Signal + SSL + phenotype	isotonic	Expected calibration error	0.0544	0.0634
Signal + SSL + phenotype	isotonic	Mean prediction error	0.0198	0.019

Table 2: Supplementary Table 7, Recalibration metrics, column block 2 of 2

model	calibration method	95% CI lower	95% CI upper	Valid bootstrap resamples
Classical signal features	raw	0.5254	0.7284	2000
Classical signal features	raw	0.2173	0.4941	2000
Classical signal features	raw	0.2234	0.2834	2000
Classical signal features	raw	-1.829	-1.014	2000
Classical signal features	raw	0.1338	0.875	2000
Classical signal features	raw	0.2102	0.3376	2000
Classical signal features	raw	0.2086	0.3371	2000
Classical signal features	platt	0.5254	0.7284	2000
Classical signal features	platt	0.2173	0.4941	2000
Classical signal features	platt	0.1266	0.1945	2000
Classical signal features	platt	-1.289	0.1958	2000
Classical signal features	platt	0.1896	1.239	2000
Classical signal features	platt	0.0341	0.1218	2000
Classical signal features	platt	-0.0427	0.0792	2000
Classical signal features	isotonic	0.5357	0.7316	2000
Classical signal features	isotonic	0.2023	0.4411	2000
Classical signal features	isotonic	0.1301	0.2044	2000
Classical signal features	isotonic	-1.395	-0.1362	2000
Classical signal features	isotonic	0.1281	0.9124	2000
Classical signal features	isotonic	0.0378	0.1342	2000
Classical signal features	isotonic	-0.0419	0.0849	2000
Signal + SSL	raw	0.4927	0.7045	2000
Signal + SSL	raw	0.2399	0.5243	2000
Signal + SSL	raw	0.2315	0.3066	2000
Signal + SSL	raw	-1.806	-0.9946	2000
Signal + SSL	raw	0.0476	0.5935	2000
Signal + SSL	raw	0.2175	0.3365	2000
Signal + SSL	raw	0.1892	0.3275	2000
Signal + SSL	platt	0.4927	0.7045	2000
Signal + SSL	platt	0.2399	0.5243	2000
Signal + SSL	platt	0.1265	0.1923	2000
Signal + SSL	platt	-1.242	0.7797	2000
Signal + SSL	platt	0.144	1.793	2000
Signal + SSL	platt	0.0298	0.1184	2000
Signal + SSL	platt	-0.0419	0.0803	2000
Signal + SSL	isotonic	0.4884	0.6902	2000
Signal + SSL	isotonic	0.17	0.3664	2000
Signal + SSL	isotonic	0.1332	0.2028	2000
Signal + SSL	isotonic	-1.577	-0.0402	2000
Signal + SSL	isotonic	-0.0321	1.026	2000
Signal + SSL	isotonic	0.027	0.1366	2000
Signal + SSL	isotonic	-0.0418	0.0828	2000

Continued from Supplementary Table 7, Recalibration metrics, column block 2 of 2.

model	calibration method	95% CI lower	95% CI upper	Valid bootstrap resamples
Signal + SSL + phenotype	raw	0.5111	0.7191	2000
Signal + SSL + phenotype	raw	0.2605	0.5471	2000
Signal + SSL + phenotype	raw	0.226	0.302	2000
Signal + SSL + phenotype	raw	-1.809	-0.9874	2000
Signal + SSL + phenotype	raw	0.0861	0.6295	2000
Signal + SSL + phenotype	raw	0.2083	0.3374	2000
Signal + SSL + phenotype	raw	0.1869	0.3227	2000
Signal + SSL + phenotype	platt	0.5111	0.7191	2000
Signal + SSL + phenotype	platt	0.2605	0.5471	2000
Signal + SSL + phenotype	platt	0.1254	0.1916	2000
Signal + SSL + phenotype	platt	-1.04	1.144	2000
Signal + SSL + phenotype	platt	0.2914	2.125	2000
Signal + SSL + phenotype	platt	0.0284	0.1098	2000
Signal + SSL + phenotype	platt	-0.0446	0.0769	2000
Signal + SSL + phenotype	isotonic	0.4914	0.6892	2000
Signal + SSL + phenotype	isotonic	0.1762	0.3749	2000
Signal + SSL + phenotype	isotonic	0.1293	0.199	2000
Signal + SSL + phenotype	isotonic	-1.459	0.161	2000
Signal + SSL + phenotype	isotonic	0.0309	1.193	2000
Signal + SSL + phenotype	isotonic	0.0199	0.113	2000
Signal + SSL + phenotype	isotonic	-0.0453	0.0784	2000

table block: recalibration differences

Table 3: Supplementary Table 7, recalibration differences, column block 1 of 2

model	metric	95% CI lower	95% CI upper	Valid bootstrap resamples
Classical signal features	brier	-0.1318	-0.0545	2000
Classical signal features	Calibration slope	0.0557	0.3639	2000
Classical signal features	Expected calibration error	-0.2573	-0.1232	2000
Classical signal features	auprc	0	0	2000
Classical signal features	auroc	0	0	2000
Classical signal features	brier	-0.1228	-0.0474	2000
Classical signal features	Calibration slope	-0.1546	0.1817	2000
Classical signal features	Expected calibration error	-0.2543	-0.1042	2000
Classical signal features	auprc	-0.0801	-0.0003	2000
Classical signal features	auroc	-0.0107	0.0311	2000
Signal + SSL	brier	-0.1551	-0.0689	2000
Signal + SSL	Calibration slope	0.0965	1.199	2000
Signal + SSL	Expected calibration error	-0.2551	-0.1422	2000
Signal + SSL	auprc	0	0	2000
Signal + SSL	auroc	0	0	2000
Signal + SSL	brier	-0.1447	-0.0631	2000
Signal + SSL	Calibration slope	-0.1522	0.4647	2000
Signal + SSL	Expected calibration error	-0.256	-0.131	2000
Signal + SSL	auprc	-0.2113	-0.0423	2000
Signal + SSL	auroc	-0.0351	0.0141	2000
Signal + SSL + phenotype	brier	-0.1516	-0.0643	2000
Signal + SSL + phenotype	Calibration slope	0.2053	1.496	2000
Signal + SSL + phenotype	Expected calibration error	-0.2588	-0.1294	2000
Signal + SSL + phenotype	auprc	0	0	2000
Signal + SSL + phenotype	auroc	0	0	2000
Signal + SSL + phenotype	brier	-0.1434	-0.0608	2000
Signal + SSL + phenotype	Calibration slope	-0.0976	0.6107	2000
Signal + SSL + phenotype	Expected calibration error	-0.263	-0.1326	2000
Signal + SSL + phenotype	auprc	-0.2366	-0.0471	2000
Signal + SSL + phenotype	auroc	-0.0606	0.01	2000

Table 4: Supplementary Table 7, recalibration differences, column block 2 of 2

model	metric	comparison	bootstrap mean difference
Classical signal features	brier	platt minus raw	-0.0937
Classical signal features	Calibration slope	platt minus raw	0.2049
Classical signal features	Expected calibration error	platt minus raw	-0.2005
Classical signal features	auprc	platt minus raw	0
Classical signal features	auroc	platt minus raw	0
Classical signal features	brier	isotonic minus raw	-0.0864
Classical signal features	Calibration slope	isotonic minus raw	0.0108
Classical signal features	Expected calibration error	isotonic minus raw	-0.19
Classical signal features	auprc	isotonic minus raw	-0.0346
Classical signal features	auroc	isotonic minus raw	0.0089
Signal + SSL	brier	platt minus raw	-0.1124
Signal + SSL	Calibration slope	platt minus raw	0.6327
Signal + SSL	Expected calibration error	platt minus raw	-0.2053
Signal + SSL	auprc	platt minus raw	0
Signal + SSL	auroc	platt minus raw	0
Signal + SSL	brier	isotonic minus raw	-0.1042
Signal + SSL	Calibration slope	isotonic minus raw	0.1199
Signal + SSL	Expected calibration error	isotonic minus raw	-0.1982
Signal + SSL	auprc	isotonic minus raw	-0.1225
Signal + SSL	auroc	isotonic minus raw	-0.0105
Signal + SSL + phenotype	brier	platt minus raw	-0.1085
Signal + SSL + phenotype	Calibration slope	platt minus raw	0.8225
Signal + SSL + phenotype	Expected calibration error	platt minus raw	-0.205
Signal + SSL + phenotype	auprc	platt minus raw	0
Signal + SSL + phenotype	auroc	platt minus raw	0
Signal + SSL + phenotype	brier	isotonic minus raw	-0.1027
Signal + SSL + phenotype	Calibration slope	isotonic minus raw	0.2035
Signal + SSL + phenotype	Expected calibration error	isotonic minus raw	-0.2065
Signal + SSL + phenotype	auprc	isotonic minus raw	-0.1351
Signal + SSL + phenotype	auroc	isotonic minus raw	-0.0263

table block: Decision curve summary

Table 5: Supplementary Table 7, Decision curve summary

threshold range	strategy	mean net benefit	Mean delta vs raw final model
0.05-0.20	Classical signal features Isotonic	0.0955	-0.0056
0.05-0.20	Classical signal features Platt	0.0992	-0.002
0.05-0.20	Classical signal features Raw	0.0944	-0.0068
0.05-0.20	Signal + SSL Isotonic	0.0885	-0.0127
0.05-0.20	Signal + SSL Platt	0.0939	-0.0073
0.05-0.20	Signal + SSL + phenotype Isotonic	0.0914	-0.0098
0.05-0.20	Signal + SSL + phenotype Platt	0.0961	-0.0051
0.05-0.20	Signal + SSL + phenotype Raw	0.1012	
0.05-0.20	Signal + SSL Raw	0.097	-0.0042
0.05-0.20	Treat all	0.0887	-0.0125
0.05-0.20	Treat none	0	-0.1012
0.10-0.30	Classical signal features Isotonic	0.0483	0.0223
0.10-0.30	Classical signal features Platt	0.045	0.019
0.10-0.30	Classical signal features Raw	0.0262	0.0001
0.10-0.30	Signal + SSL Isotonic	0.0376	0.0115
0.10-0.30	Signal + SSL Platt	0.0405	0.0144
0.10-0.30	Signal + SSL + phenotype Isotonic	0.0389	0.0128
0.10-0.30	Signal + SSL + phenotype Platt	0.0442	0.0181
0.10-0.30	Signal + SSL + phenotype Raw	0.0261	
0.10-0.30	Signal + SSL Raw	0.0263	0.0002
0.10-0.30	Treat all	0.0003	-0.0258
0.10-0.30	Treat none	0	-0.0261
0.05-0.50	Classical signal features Isotonic	0.0326	0.0671
0.05-0.50	Classical signal features Platt	0.0362	0.0708
0.05-0.50	Classical signal features Raw	-0.0397	-0.0051
0.05-0.50	Signal + SSL Isotonic	0.0254	0.0599
0.05-0.50	Signal + SSL Platt	0.0389	0.0734
0.05-0.50	Signal + SSL + phenotype Isotonic	0.0311	0.0656
0.05-0.50	Signal + SSL + phenotype Platt	0.0405	0.075
0.05-0.50	Signal + SSL + phenotype Raw	-0.0345	
0.05-0.50	Signal + SSL Raw	-0.0385	-0.004
0.05-0.50	Treat all	-0.136	-0.1015
0.05-0.50	Treat none	0	0.0345